

Azure Data Factory – Data Flow Activity

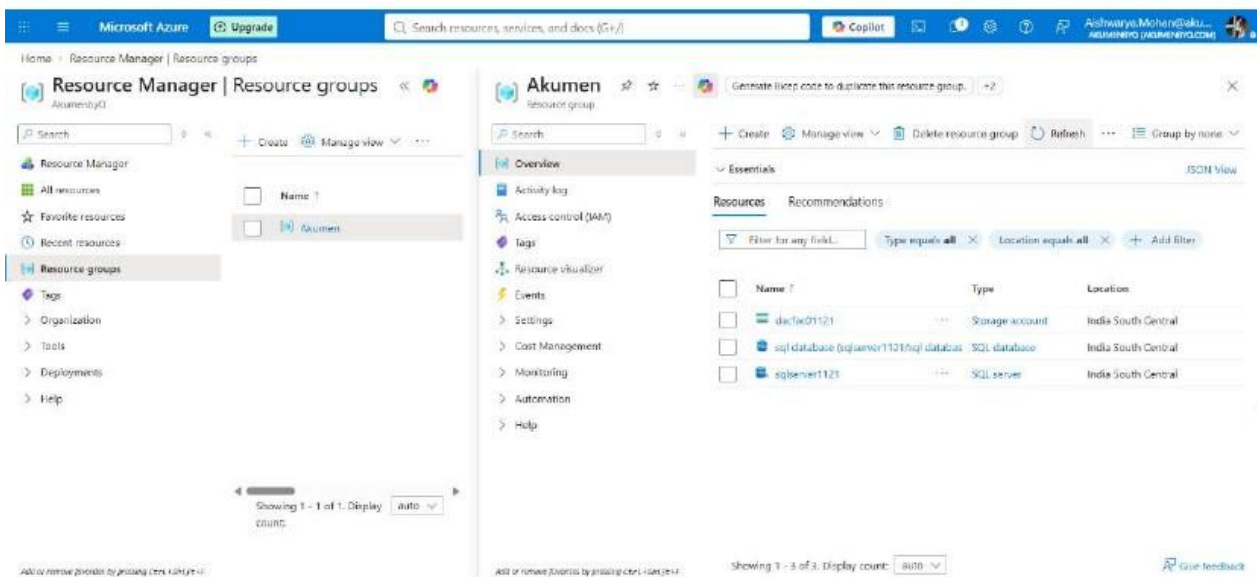
Filter + Sort | Join → Sink to Blob

This document shows the complete end-to-end process in Azure:

1. Create Resource Group, Storage Account (input/output containers)
2. Create Azure SQL Server + Database + sample Employee table
3. Create Azure Data Factory + Linked Services (SQL + Blob)
4. Data Flow 1: Source (SQL) → Filter (IT) → Sort → Sink (Blob)
5. Data Flow 2: Two Sources → Join → Sink (Blob)

① Resource Group & Storage Account

Resource group 'Akumen' created → contains Storage Account + SQL resources

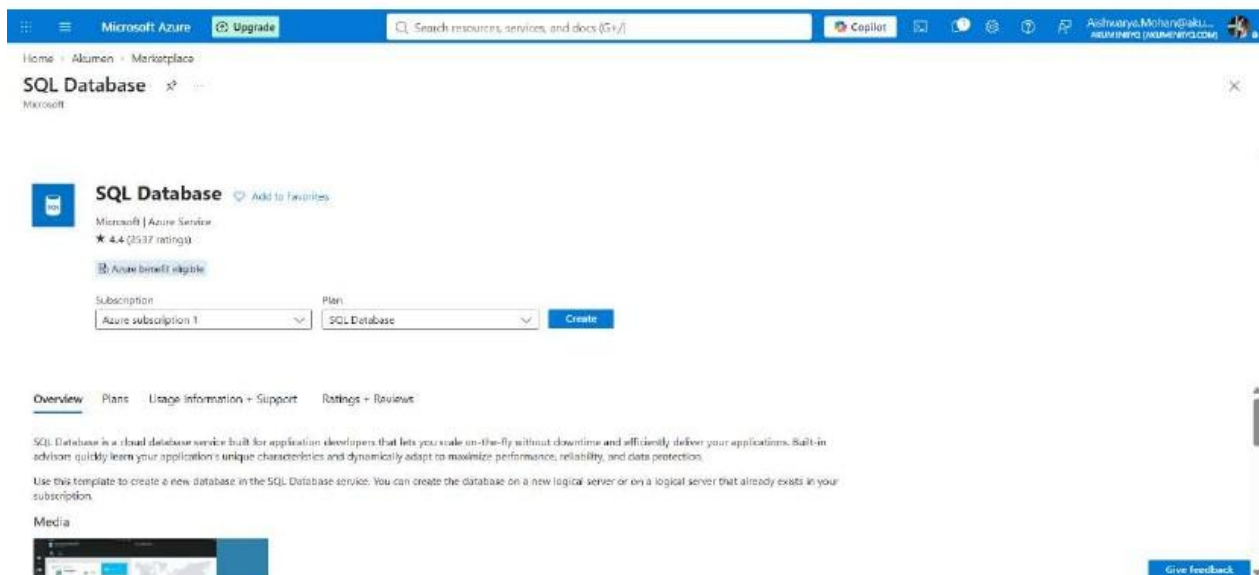


Storage account containers: input + output

ADF Data Flow – Screenshot Project

② Create Azure SQL Database

Marketplace → SQL Database selected



Create SQL Database → Server name & authentication

② Create Azure SQL Database (continued)

Server admin login + password set → Use both SQL and Microsoft Entra authentication

Microsoft Azure Upgrade Search resources, services, and docs (51/1) Copilot Ashwarya.Mohar@akumenbyq.com

Home > Create SQL Database

Create SQL Database Server

Microsoft

Info Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#) ⓘ

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. [Learn more](#) about using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. [Learn more](#) ⓘ

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☒ Use both SQL and Microsoft Entra authentication
- ☐ Use SQL authentication

Set Microsoft Entra admin

Admin Object/App ID: 23a262f7-470a-4391-9057-6339f4ab8527
[Set admin](#)

Server admin login * ash1121 ✓

Password * [REDACTED] ✓

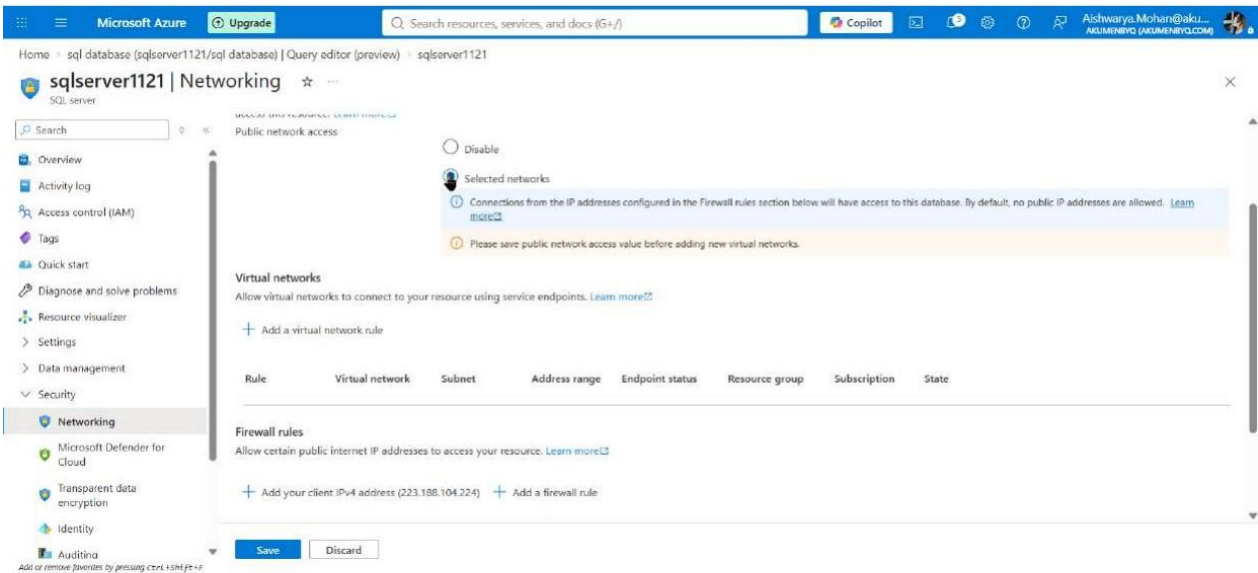
Confirm password * [REDACTED] ✓

OK Feedback

Deployment succeeded

③ Enable Access & Open Query Editor

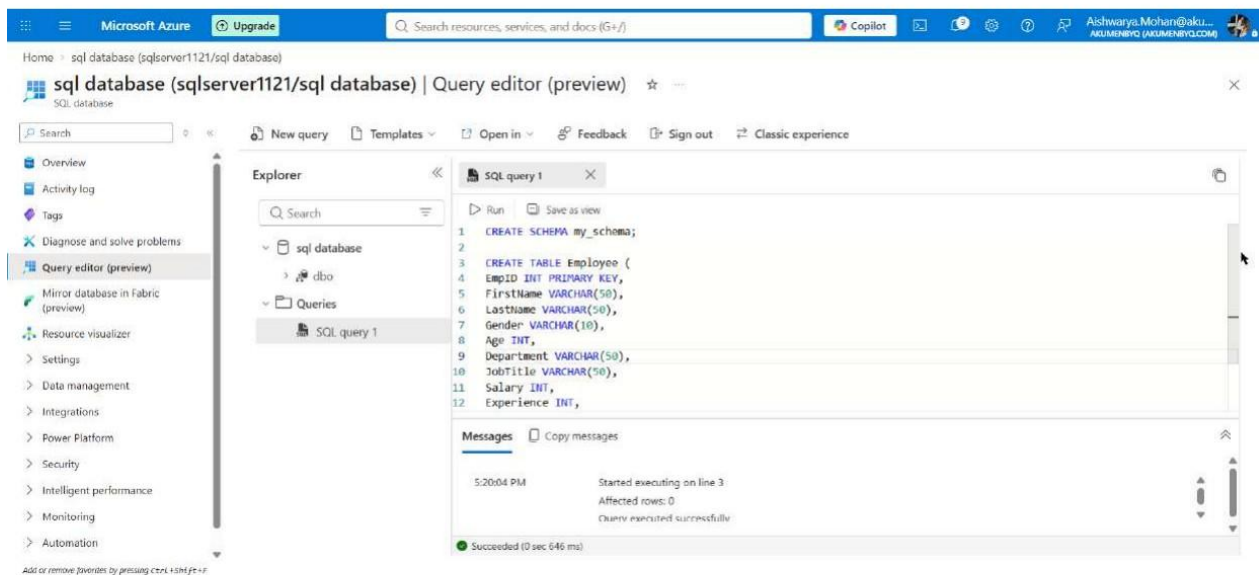
Networking → Allow Azure services + add client IP



Query Editor → Connect with SQL authentication

④ Create Employee Table & Insert Data

CREATE TABLE Employee ... executed successfully



INSERT multiple rows → data loaded

④ Verify Data & Optional View

SELECT * FROM Employee → data preview

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/f) | Copilot | Aishwarya.Mohan@aku... | AKUMH@PVO (AKUMH@PVO.COM)

Home > sql database (sqlserver1121/sql database) | Query editor (preview) | SQL database

Search | New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- sql database
 - dbo
 - Tables
 - Employee

SQL query 1

```
43 INSERT INTO Employee (EmpID, FirstName, LastName, Gender, Age, Department, JobTitle, Salary, Experience)
44 VALUES (25, 'Sanjay', 'Puri', 'Male', 45, 'Sales', 'Regional Sales Head', 75000, 18),
45 (26, 'Julie', 'Dacruz', 'Female', 26, 'Marketing', 'Digital Marketer', 39000, 2),
46 (27, 'Rohan', 'Kulkarni', 'Male', 31, 'IT', 'DevOps Engineer', 61000, 6),
47 (28, 'Tanya', 'Shetty', 'Female', 34, 'Finance', 'Risk Analyst', 64000, 0),
48 (29, 'Akash', 'Halli', 'Male', 23, 'HR', 'HR Intern', 22000, 0),
49 (30, 'Reema', 'Jain', 'Female', 29, 'IT', 'QA Engineer', 50000, 5);
50 select * from Employee;
```

Messages | Results

	EmpID	FirstName	LastName	Gender	Age	Department	JobTitle	Salary	Experience
1	1	Rahul	Sharma	Male	28	IT	Software Engi...	55000	3
2	2	Priya	Mehra	Female	32	HR	HR Manager	65000	7
3	3	Amit	Verma	Male	45	Finance	Accountant	48000	15
4	4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4
5	5	Karan	Singh	Male	35	Sales	Sales Executive	40000	8

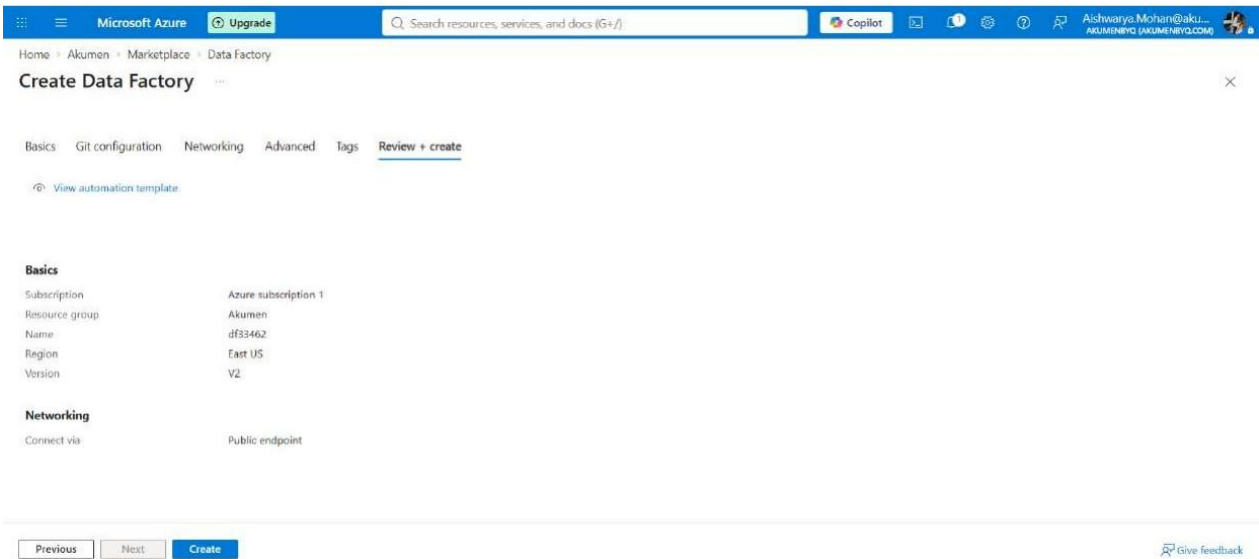
Succeeded (0 sec 204 ms) | Columns: 12 Rows: 30

Add or remove favorites by pressing ctrl+shift+F

CREATE VIEW EmployeeData (optional)

⑤ Create Azure Data Factory

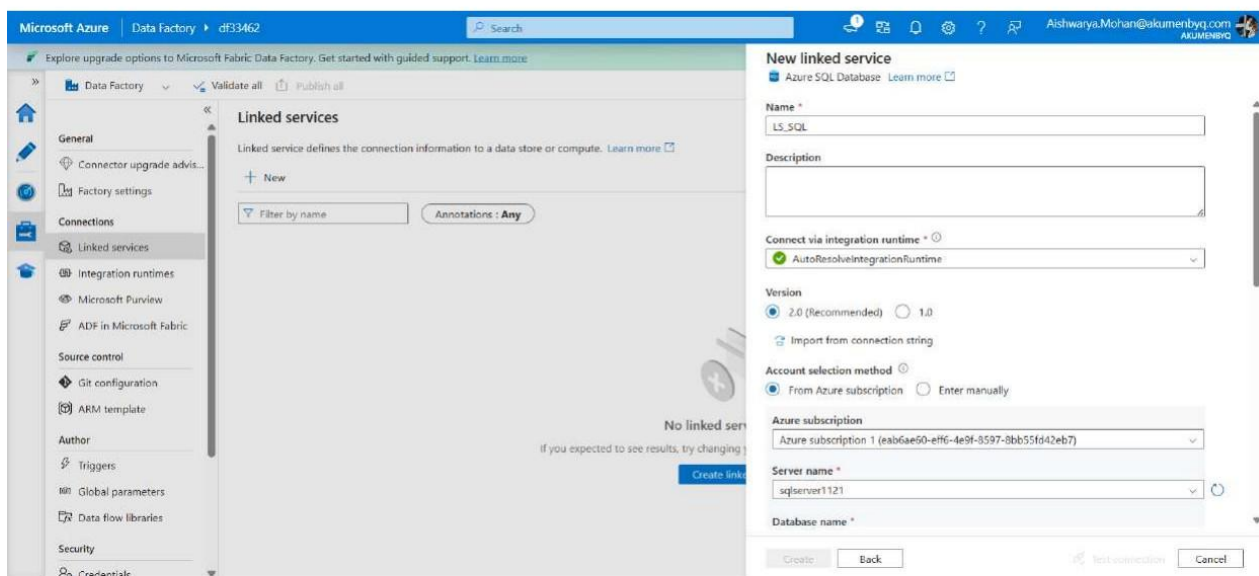
Create Data Factory → Review + create



Resource group now shows Data Factory + SQL + Storage

⑥ Create Linked Services

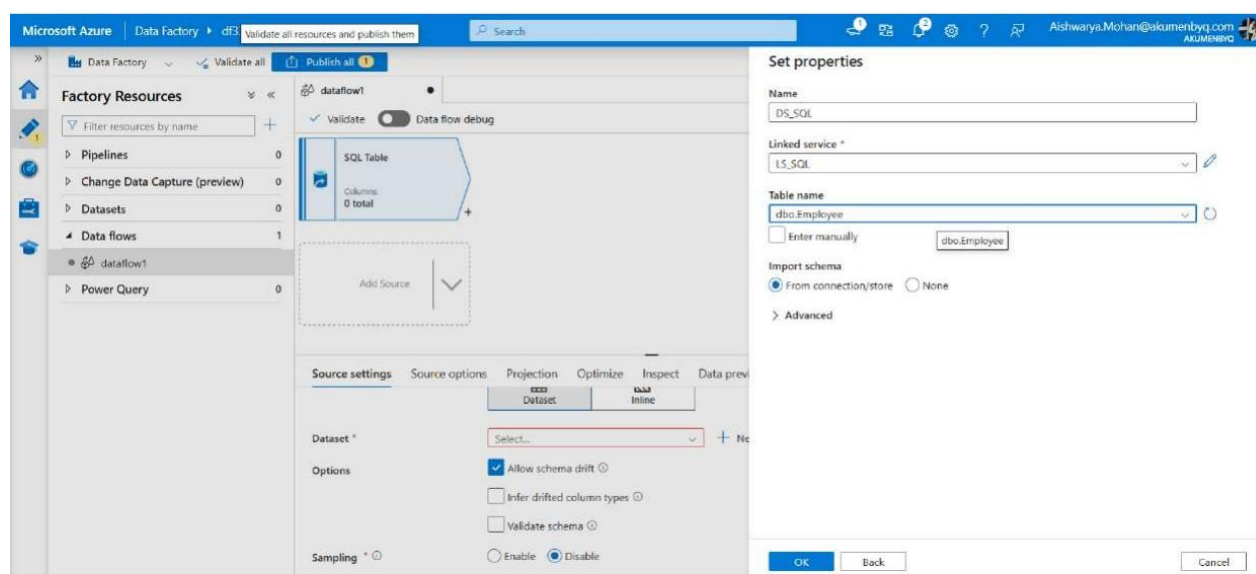
New linked service → Azure SQL Database (LS_SQL)



New linked service → Azure Blob Storage (LS_Blob) also created

⑦ Data Flow 1 – Add Source (SQL)

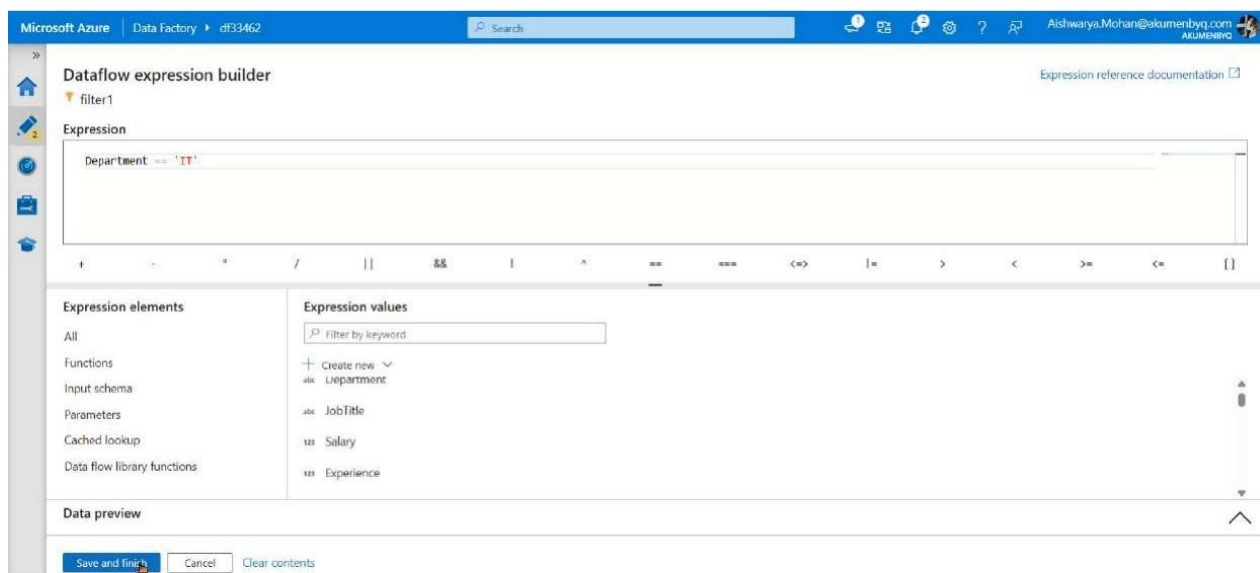
New Data Flow → Add Source → Azure SQL Dataset



Source settings → Table = Employee → Import schema

⑧ Data Flow 1 – Filter (Department = 'IT')

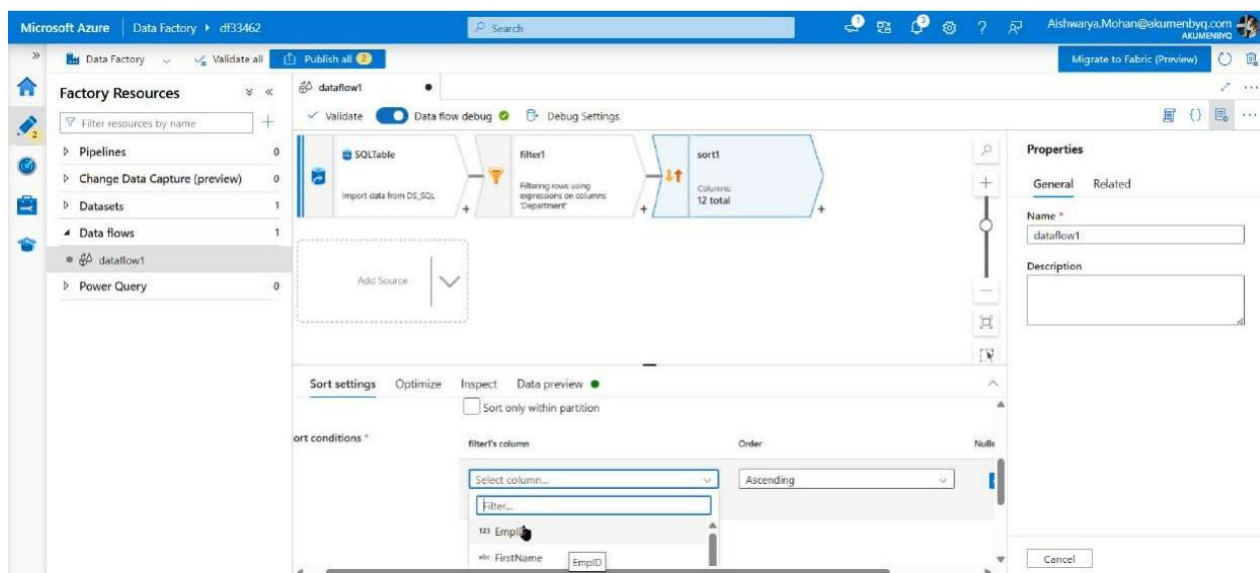
Add Filter transformation → Expression: Department == 'IT'



Data preview after Filter → only IT rows

⑨ Data Flow 1 – Sort (Ascending)

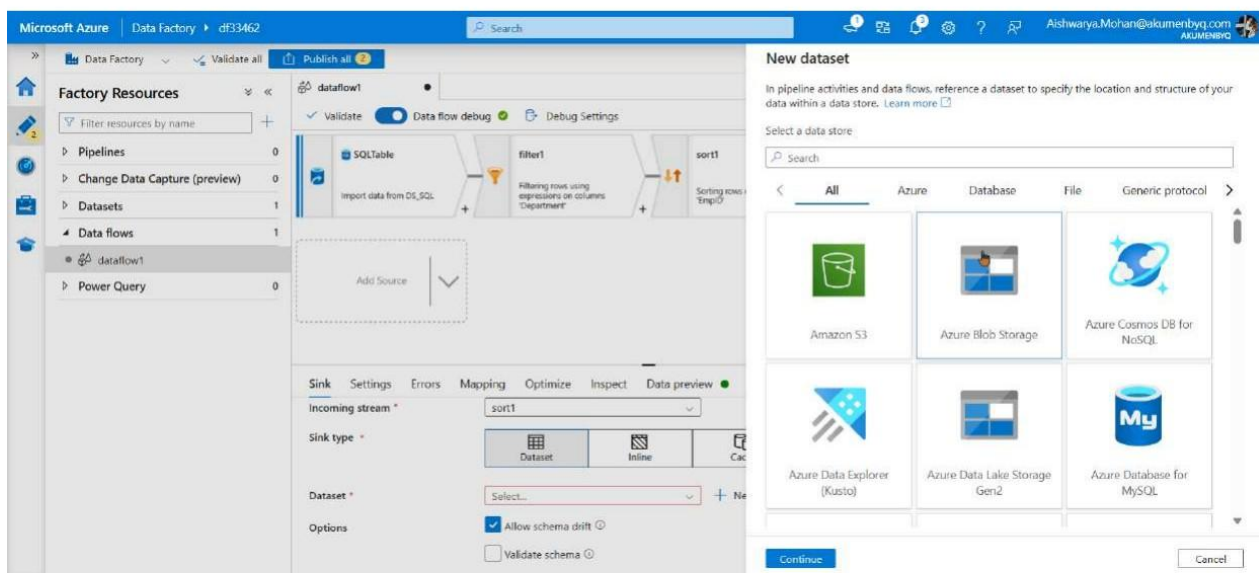
Add Sort transformation → Sort by EmpID / FirstName Ascending



Data preview after Sort

⑩ Data Flow 1 – Sink to Blob

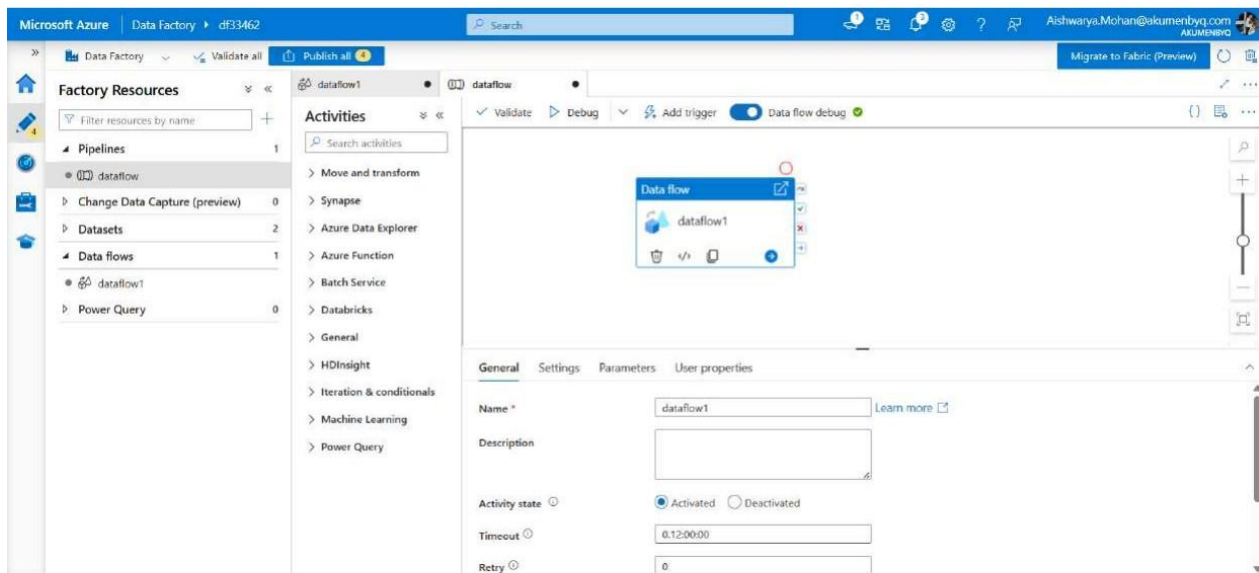
Add Sink → Dataset pointing to output container



Sink configured → Test connection successful

Pipeline → Debug Run (Part 1)

Create Pipeline → Add Data Flow activity





Debug run → Succeeded

Verify Output in Blob Storage

output container → CSV file generated



Microsoft Azure | Upgrade | Search resources, services, and docs (G+/I) | Copilot | Aishwarya.Mohan@aku... AKU14282970 (AKU14282970.COM)

Home > dacfac01121 | Containers

output
Container

Search | + Add Directory

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method

+ Add filter

Search blobs by

Showing all 2 items

☐ Name

☐ _SUCCESS

☒ part-00

part-00000-d98d4100-b058-455d-aee5-cec192c3ec0a-c000.csv
Blob

Save | Discard | Download | Refresh | Delete

Overview | Versions | Snapshots | **Edit** | Generate SAS

	EmpID	FirstName	LastName	Gender	Age	Department	JobTitle	Salary	Experience	City	ManagerID	JoinDate
1	1	Rahul	Sharma	Male	28	IT	Software Engineer	55000	3	Mumbai	101	2021-05-10
2	4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4	Bangalore	101	2020-08-20
3	7	Rohit	Mair	Male	41	IT	Team Lead	72000	12	Delhi	101	2012-09-10
4	11	Suresh	Menon	Male	38	IT	System Admin	50000	10	Bangalore	101	2014-07-19
5	14	Kavya	Shah	Female	28	IT	Backend Developer	56000	4	Pune	101	2019-06-15
6	18	Ritika	Soni	Female	27	IT	Frontend Developer	54000	3	Bangalore	101	2021-03-09
7	21	Lokesh	Bhat	Male	39	IT	Cloud Engineer	68000	11	Hyderabad	101	2014-02-17
8	27	Rohan	Kulkarni	Male	31	IT	DevOps Engineer	61000	6	Bangalore	101	2017-05-29
9	30	Reema	Jain	Female	29	IT	QA Engineer	50000	5	Chennai	101	2019-12-19
10												
11												

Csv | Preview

Add or remove favorites by pressing ctrl + shift + F

Final filtered & sorted data written to Blob

Part 2 – Create Departments Table

CREATE TABLE Departments + INSERT data

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/f) | Copilot | Aishwarya.Mohan@aku... AKUMENINGO (AKUMENINGO.COM)

Home > sql database (sqlserver1121/sql database) | Query editor (preview) ☆

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation

Explorer

sql database
dbo
Tables
Departments
Employee
Views
Employee...
Stored Proced...
Functions
Queries
SQL query 1

SQL query 2

```
5 );  
6 Insert into Departments values  
7 (1,'Data Engineering','Bangalore'),  
8 (2,'Analytics','Hyderabad'),  
9 (3,'Data Science','Bangalore'),  
10 (4,'IT Support','Chennai');  
11 select * from Departments;  
12
```

Messages Results

	DeptD	DeptName	Location
1	1	Data Engineering	Bangalore
2	2	Analytics	Hyderabad
3	3	Data Science	Bangalore
4	4	IT Support	Chennai

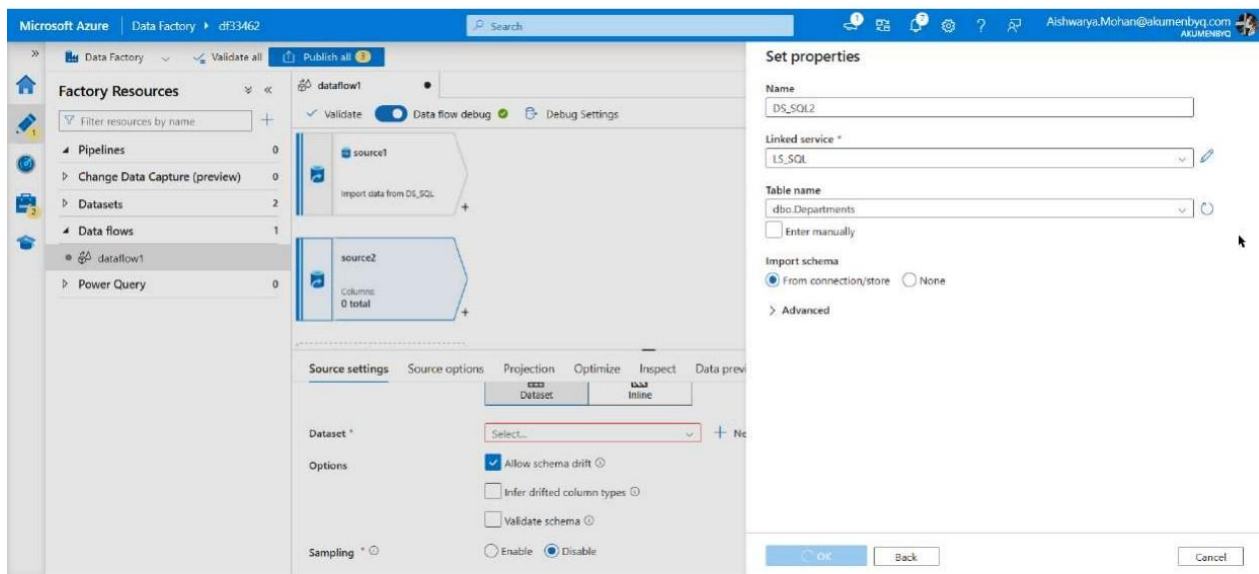
Succeeded (0 sec 171 ms) Columns: 3 Rows: 4



Departments table ready for Join

Data Flow 2 – Two Sources + Join

New Data Flow → source1 (Employee) + source2 (Departments)

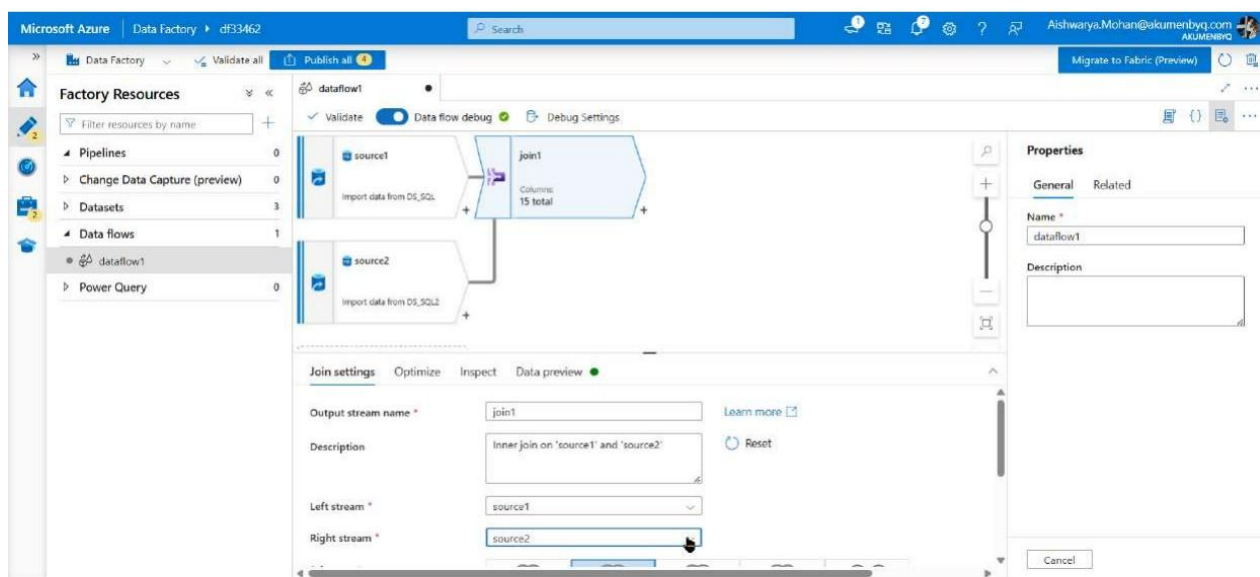




Add Join transformation

Configure Join & Preview

Join settings → Left = Employee, Right = Departments, Key match

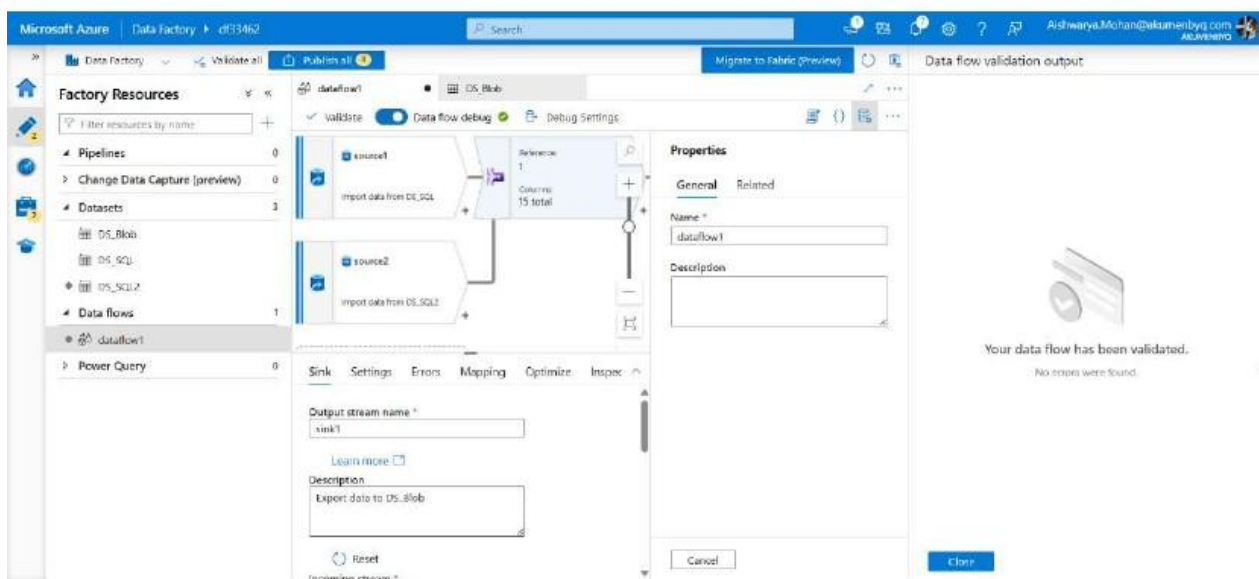




Data preview after Join

Sink + Pipeline Run (Part 2)

Add Sink → same Blob output container





Pipeline debug → Data Flow succeeded

Overall Logic

Part 1 – Filter + Sort

Source (SQL Employee table) → Filter (Department = 'IT') → Sort (Ascending) → Sink (Blob output)

Part 2 – Join

Source1 (Employee) + Source2 (Departments) → Join → Sink (Blob output)

Prerequisites completed before Data Factory:

- Resource Group: Akumen
- Storage Account with containers: input / output
- Azure SQL Server + Database with Employee (and Departments) table
- Linked Services: LS_SQL (Azure SQL Database) and LS_Blob (Azure Blob Storage)